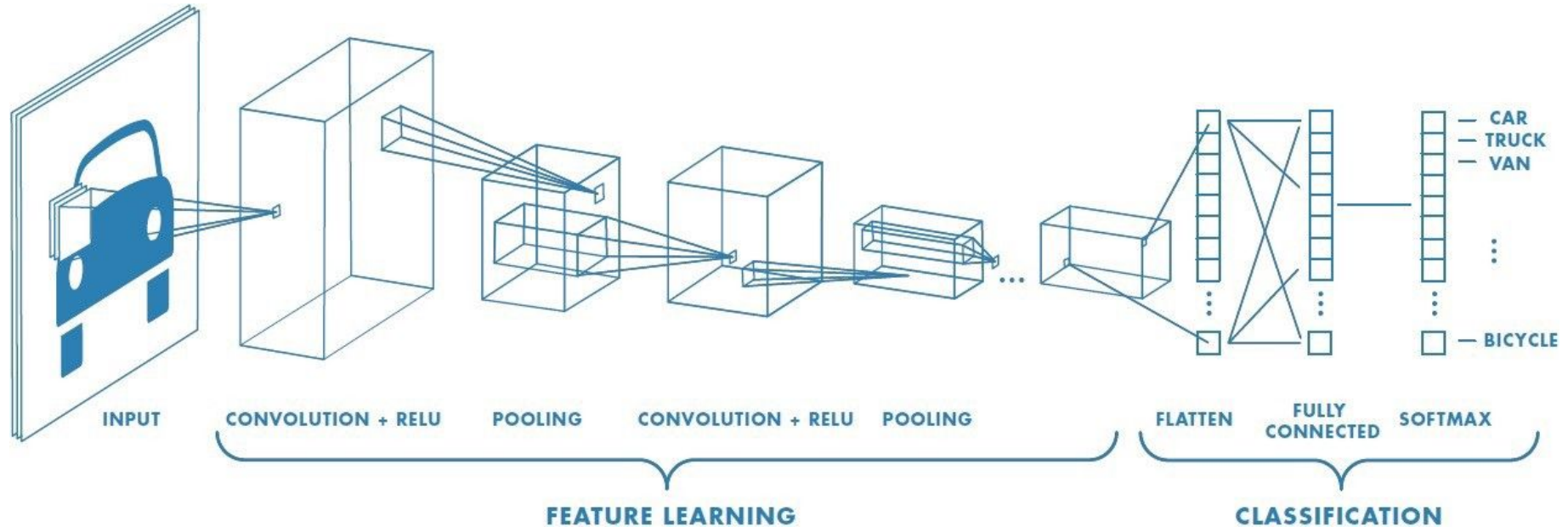
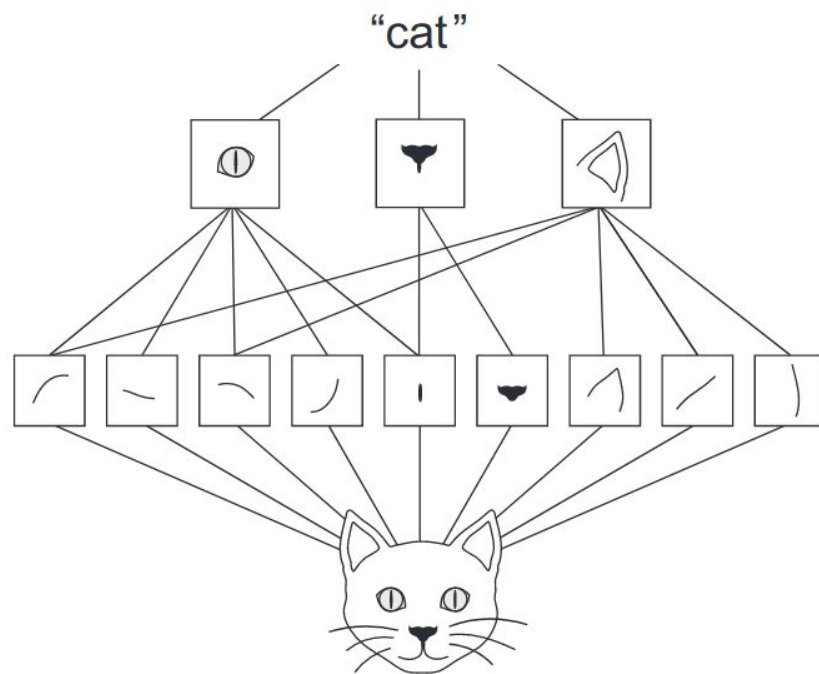




Convolutional Neural Networks (CNNs)

Convolutional Neural Networks (CNNs)



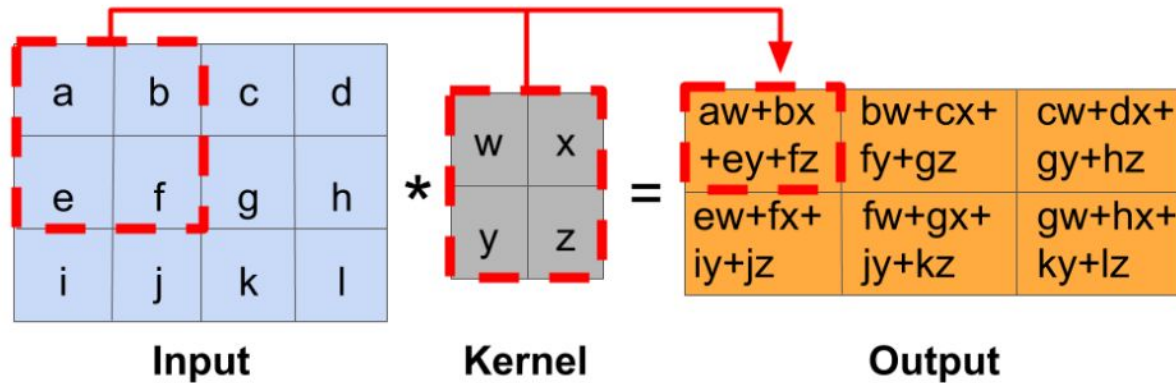




CNNs – Kernel

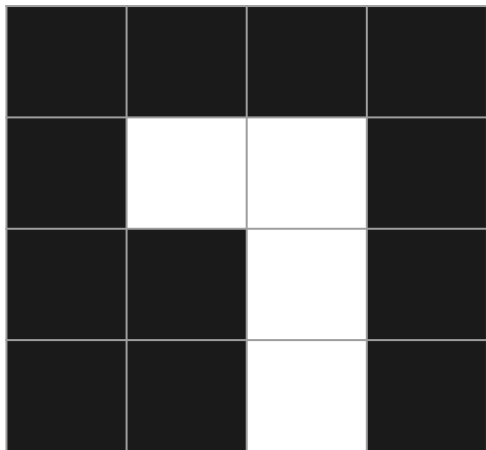
- Also known as Filter.
- Small matrices of weights (coefficients).
- They slide across the input data (like an image), performing element-wise operations.
- They learn specific features.

Convolution Process



Fuente: Bengio, Goodfellow y Courville (2016)

Convolution Process



0	0	0	0
0	1	1	0
0	0	1	0
0	0	1	0

Convolution Process



Input (4x4)

0	0	0	0
0	1	1	0
0	0	1	0
0	0	1	0

Kernel (2x2)

1	1
0	1

Convolution Process

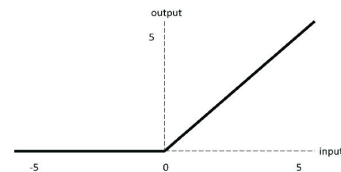
Input (4x4)

0	0	0	0
0	1	1	0
0	0	1	0
0	0	1	0

Kernel (2x2)

1	1
0	1

ReLU



Output (3x3)

1		

Convolution Process

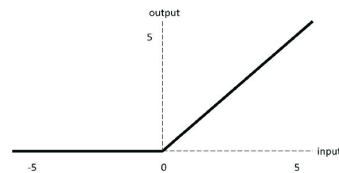
Input (4x4)

0	0	0	0
0	1	1	0
0	0	1	0
0	0	1	0

Kernel (2x2)

1	1
0	1

ReLU



Output (3x3)

1	1	

Convolution Process

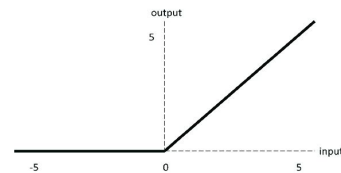
Input (4x4)

0	0	0	0
0	1	1	0
0	0	1	0
0	0	1	0

Kernel (2x2)

1	1
0	1

ReLU



Output (3x3)

1	1	0

Convolution Process

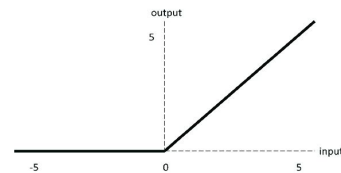
Input (4x4)

0	0	0	0
0	1	1	0
0	0	1	0
0	0	1	0

Kernel (2x2)

1	1
0	1

ReLU



Output (3x3)

1	1	0
1		

Convolution Process

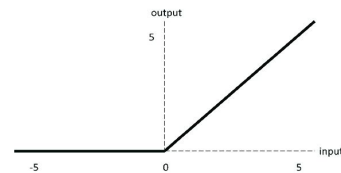
Input (4x4)

0	0	0	0
0	1	1	0
0	0	1	0
0	0	1	0

Kernel (2x2)

1	1
0	1

ReLU



Output (3x3)

1	1	0
1	3	

Convolution Process

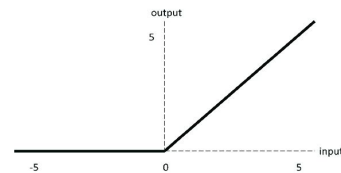
Input (4x4)

0	0	0	0
0	1	1	0
0	0	1	0
0	0	1	0

Kernel (2x2)

1	1
0	1

ReLU



Output (3x3)

1	1	0
1	3	1

Convolution Process

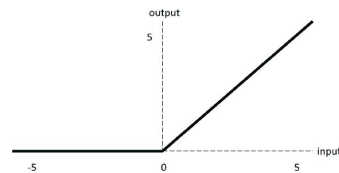
Input (4x4)

0	0	0	0
0	1	1	0
0	0	1	0
0	0	1	0

Kernel (2x2)

1	1
0	1

ReLU



Output (3x3)

1	1	0
1	3	1
0		

Convolution Process

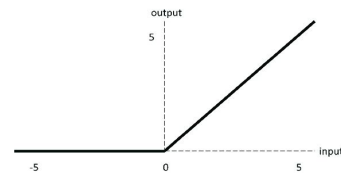
Input (4x4)

0	0	0	0
0	1	1	0
0	0	1	0
0	0	1	0

Kernel (2x2)

1	1
0	1

ReLU



Output (3x3)

1	1	0
1	3	1
0	2	

Convolution Process

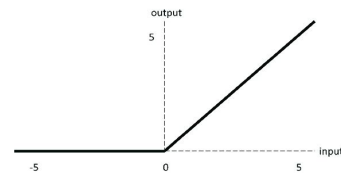
Input (4x4)

0	0	0	0
0	1	1	0
0	0	1	0
0	0	1	0

Kernel (2x2)

1	1
0	1

ReLU



Output (3x3)

1	1	0
1	3	1
0	2	1

Convolution Process

Input (4x4)

0	0	0	0
0	1	1	0
0	0	1	0
0	0	1	0

Kernel (2x2)

1	1
0	1

Output (3x3)

1	1	0
1	3	1
0	2	1



CNNs – Pooling

- Technique to reduce the spatial dimensions (e.g., of an image).
 - Reduces computational cost.
 - Makes the network more robust, reducing sensibility to small variations.
 - Prevents overfitting by reducing the number of parameters.
- Most common pooling layers:
 - Max
 - Average

Max-Pooling (2x2)

Input (3x3)
(from last convolution)

1	1	0
1	3	1
0	2	1

Output (2x2)

Max-Pooling (2x2)

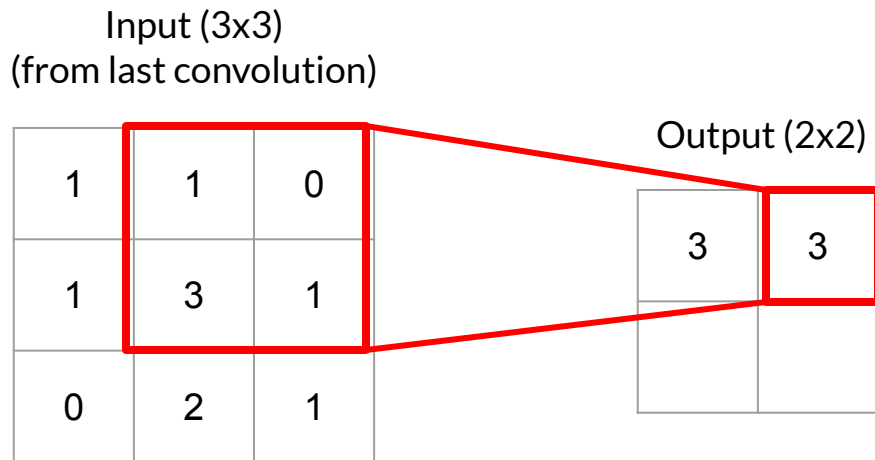
Input (3x3)
(from last convolution)

1	1	0
1	3	1
0	2	1

Output (2x2)

3	

Max-Pooling (2x2)



Max-Pooling (2x2)

Input (3x3)
(from last convolution)

1	1	0
1	3	1
0	2	1

Output (2x2)

3	3
3	

Max-Pooling (2x2)

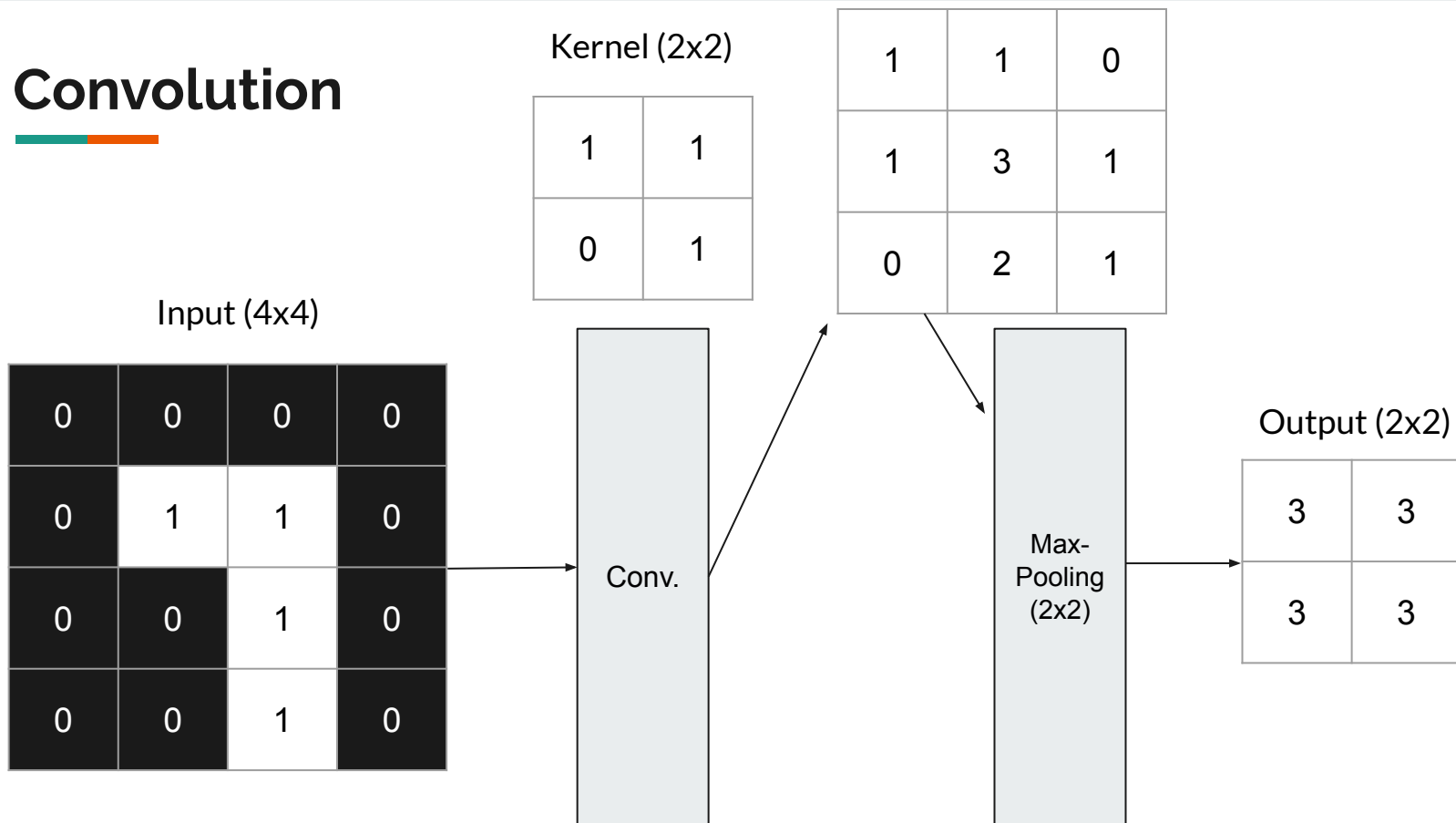
Input (3x3)
(from last convolution)

1	1	0
1	3	1
0	2	1

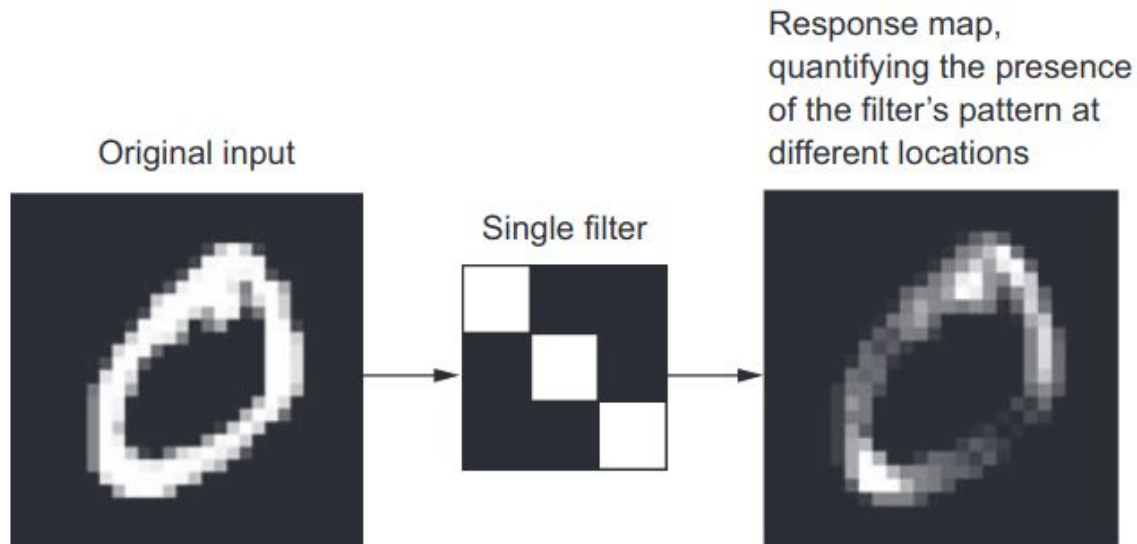
Output (2x2)

3	3
3	3

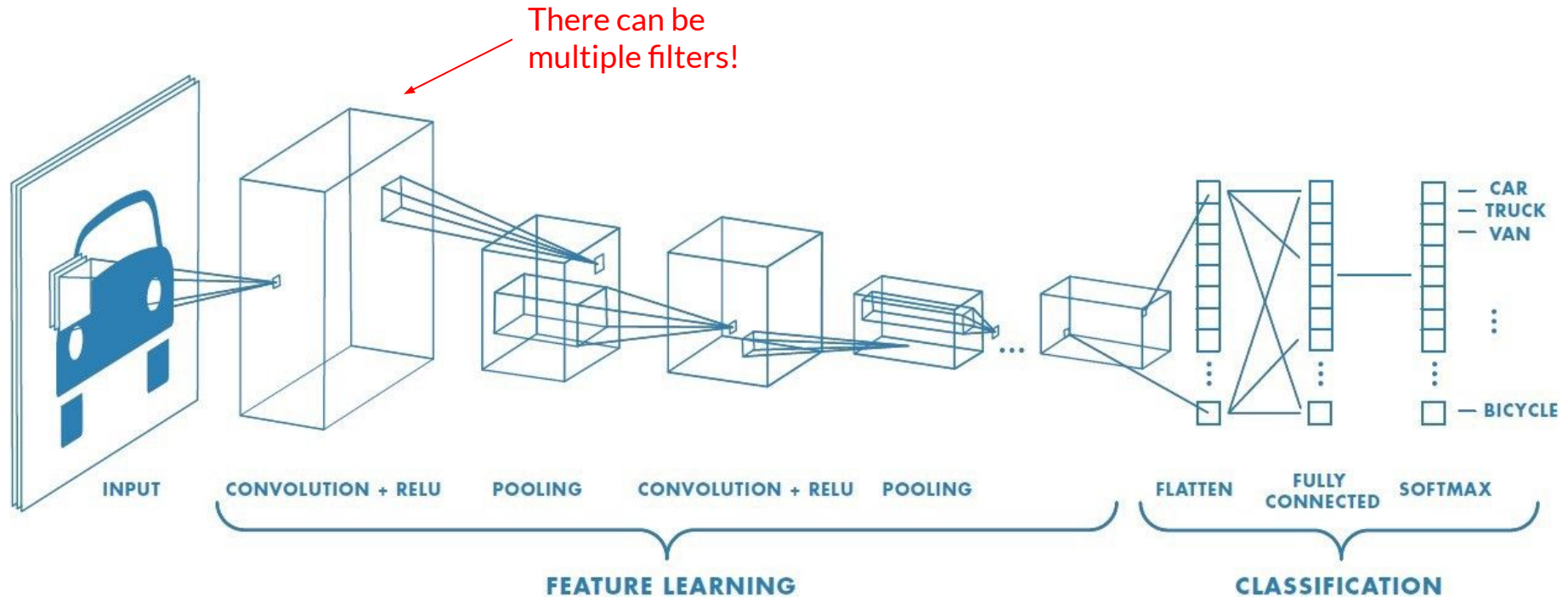
Convolution



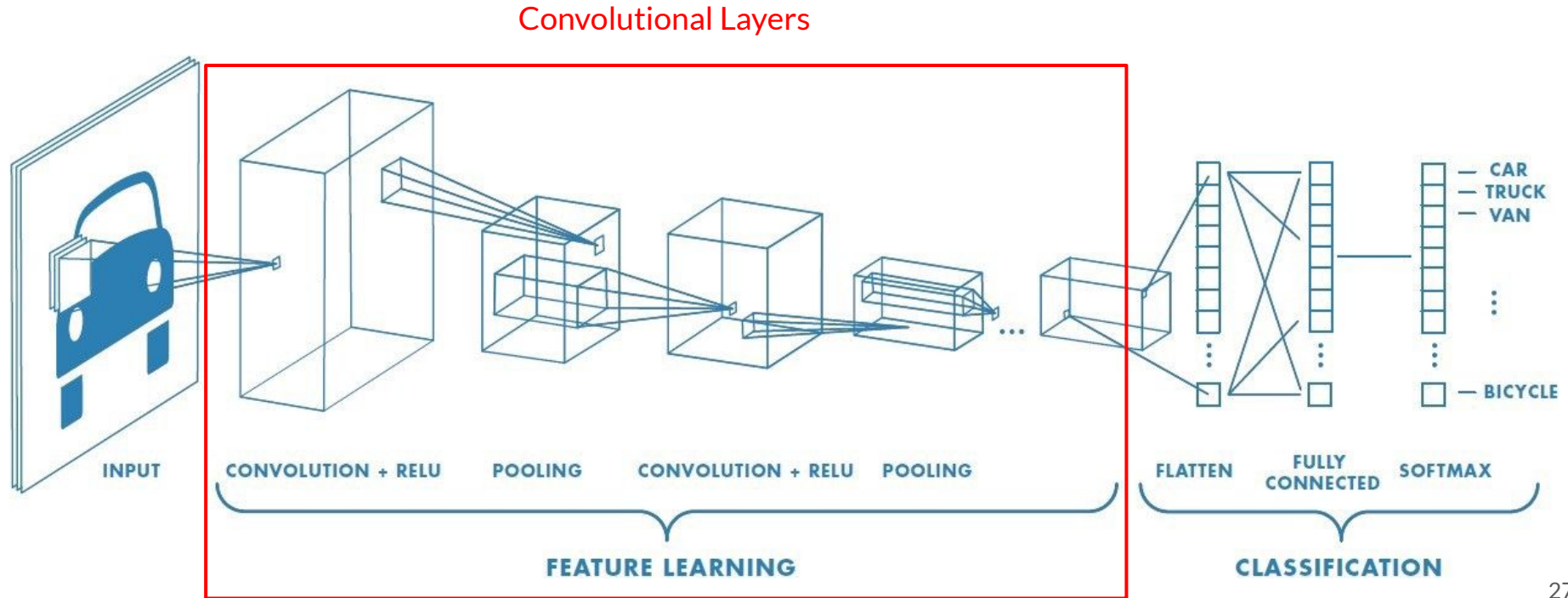
Convolution



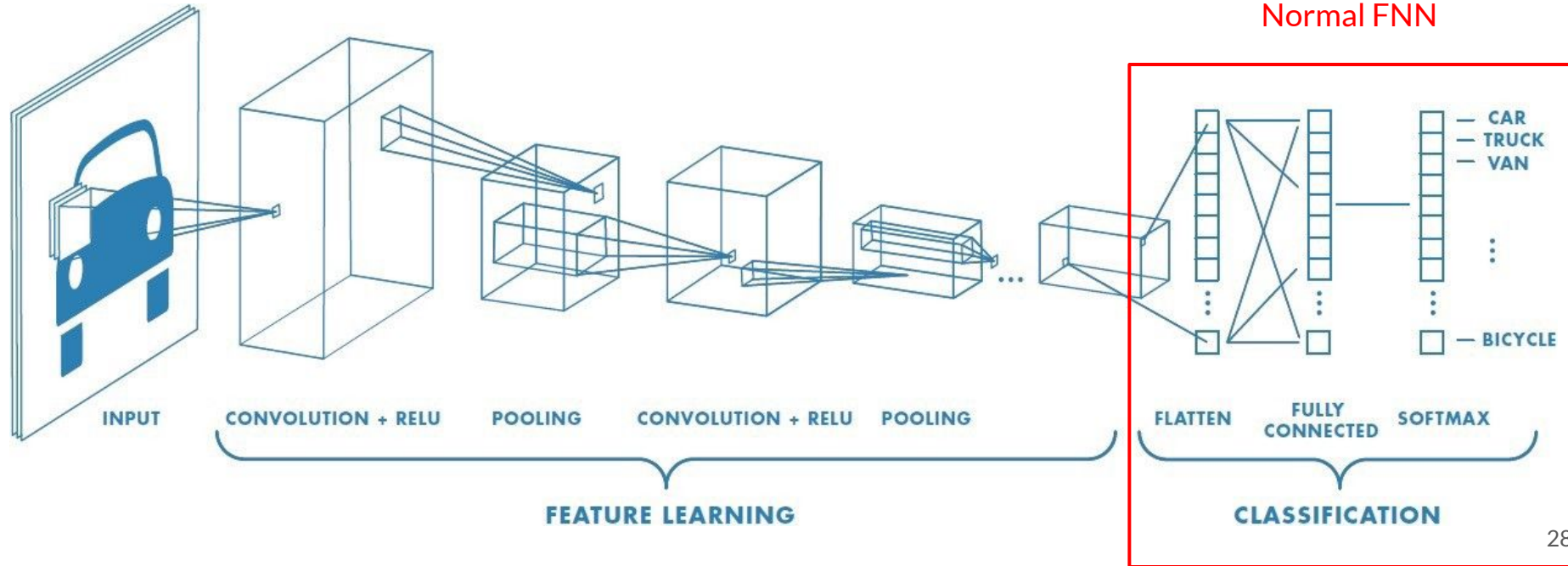
Convolutional Neural Networks (CNNs)



Convolutional Neural Networks (CNNs)



Convolutional Neural Networks (CNNs)

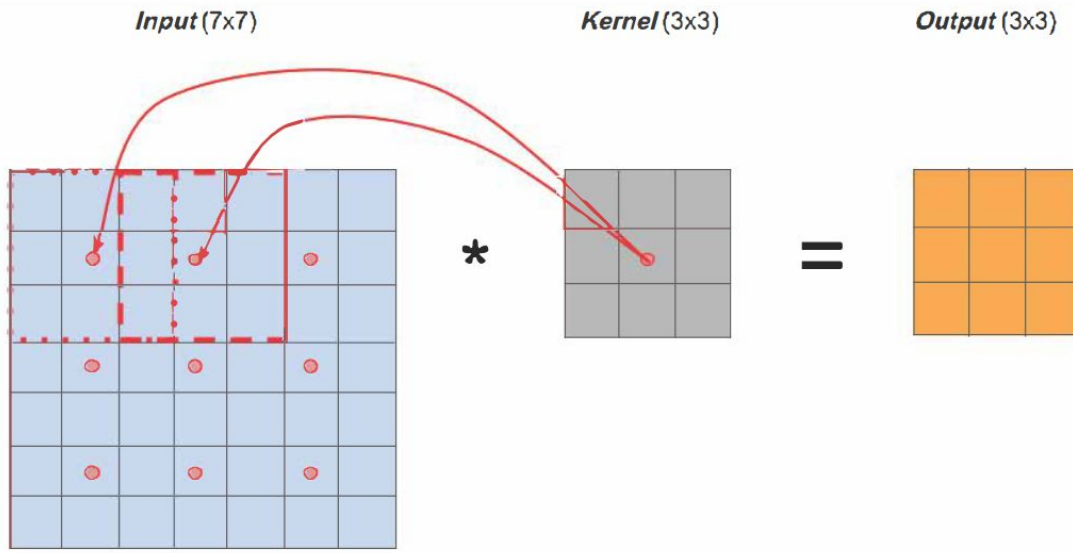




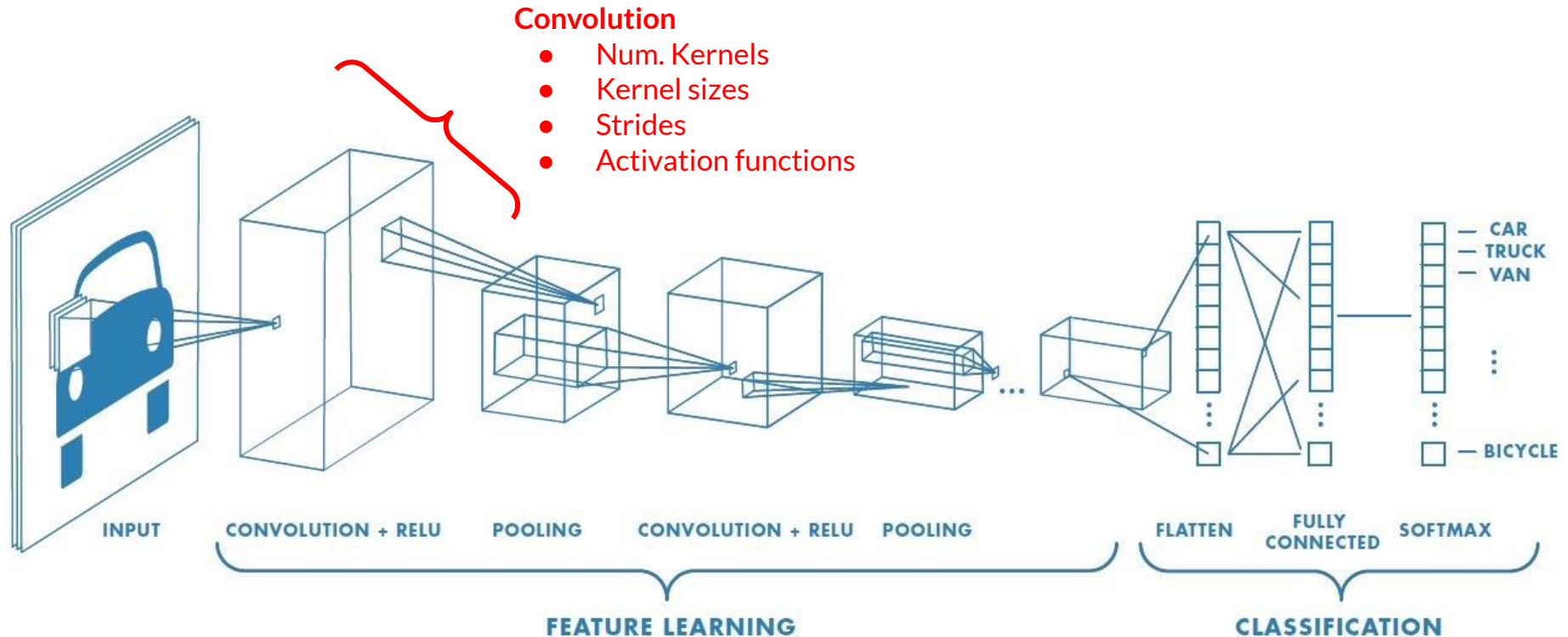
Convolution

https://adamharley.com/nn_vis/cnn/2d.html

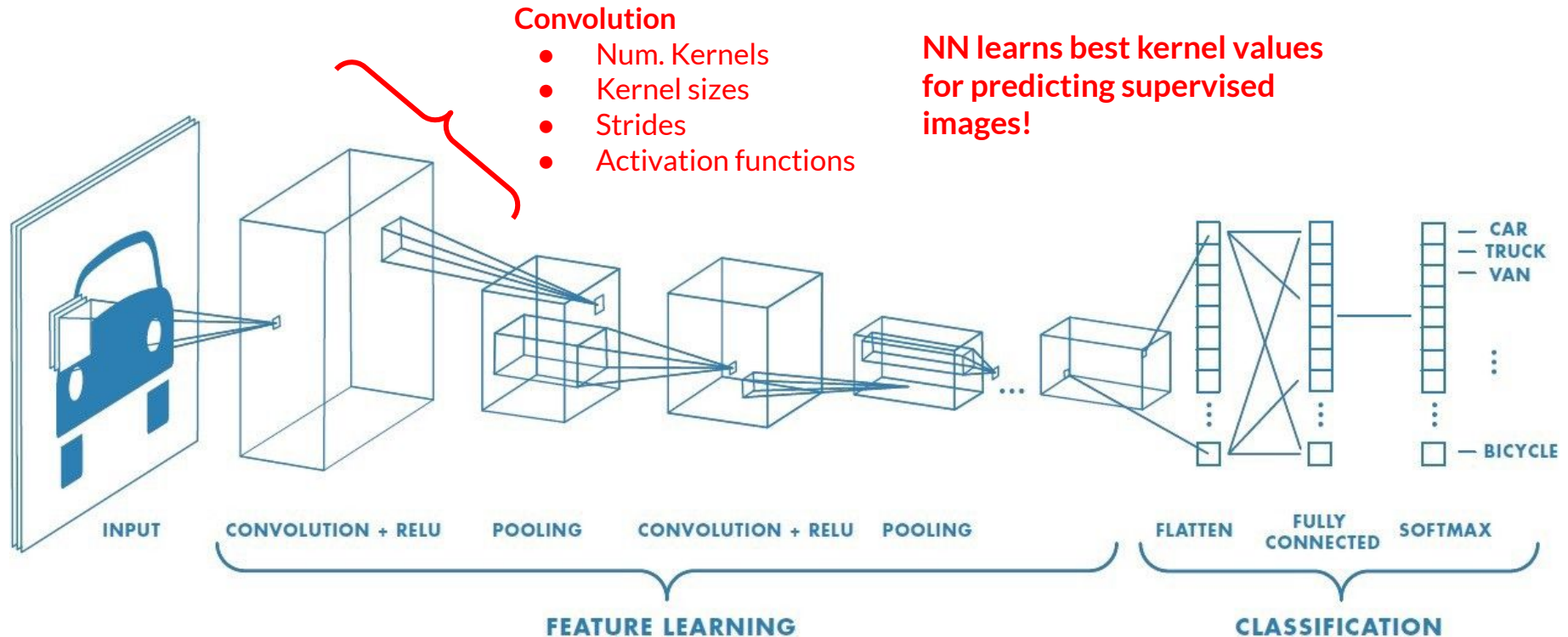
Stride = 2



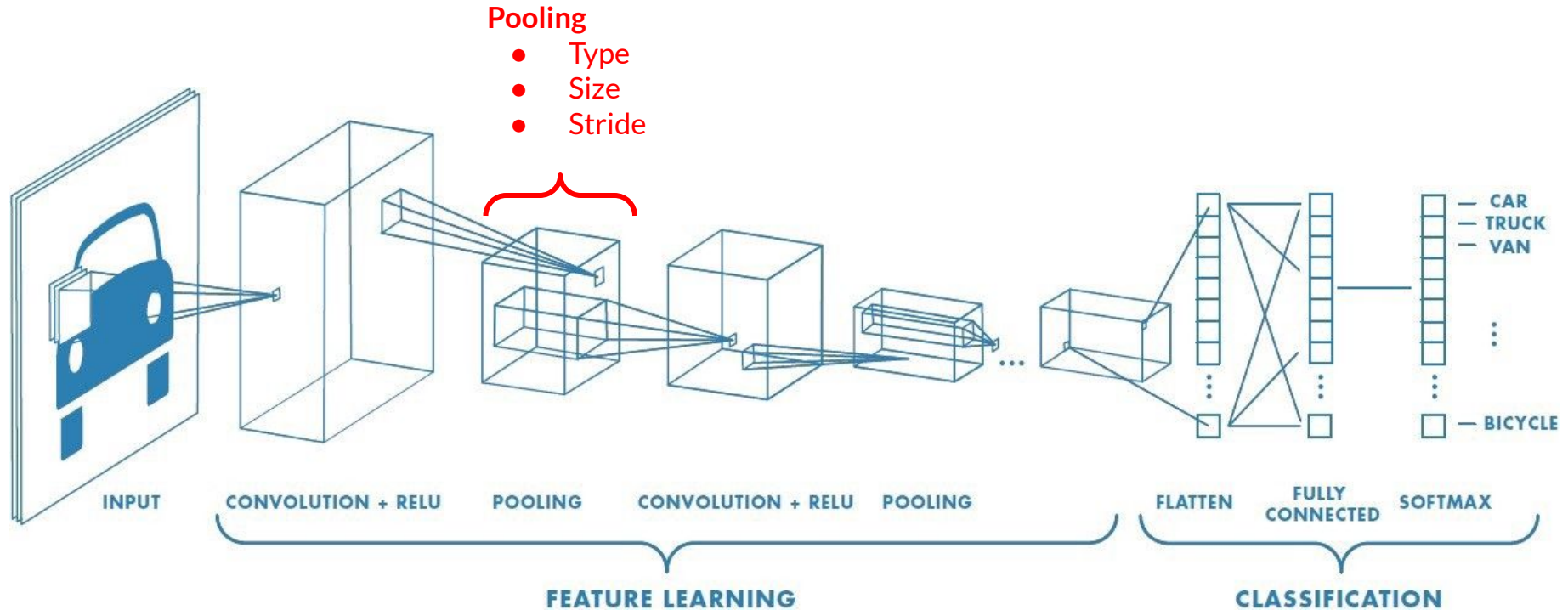
Convolutional Neural Networks (CNNs)



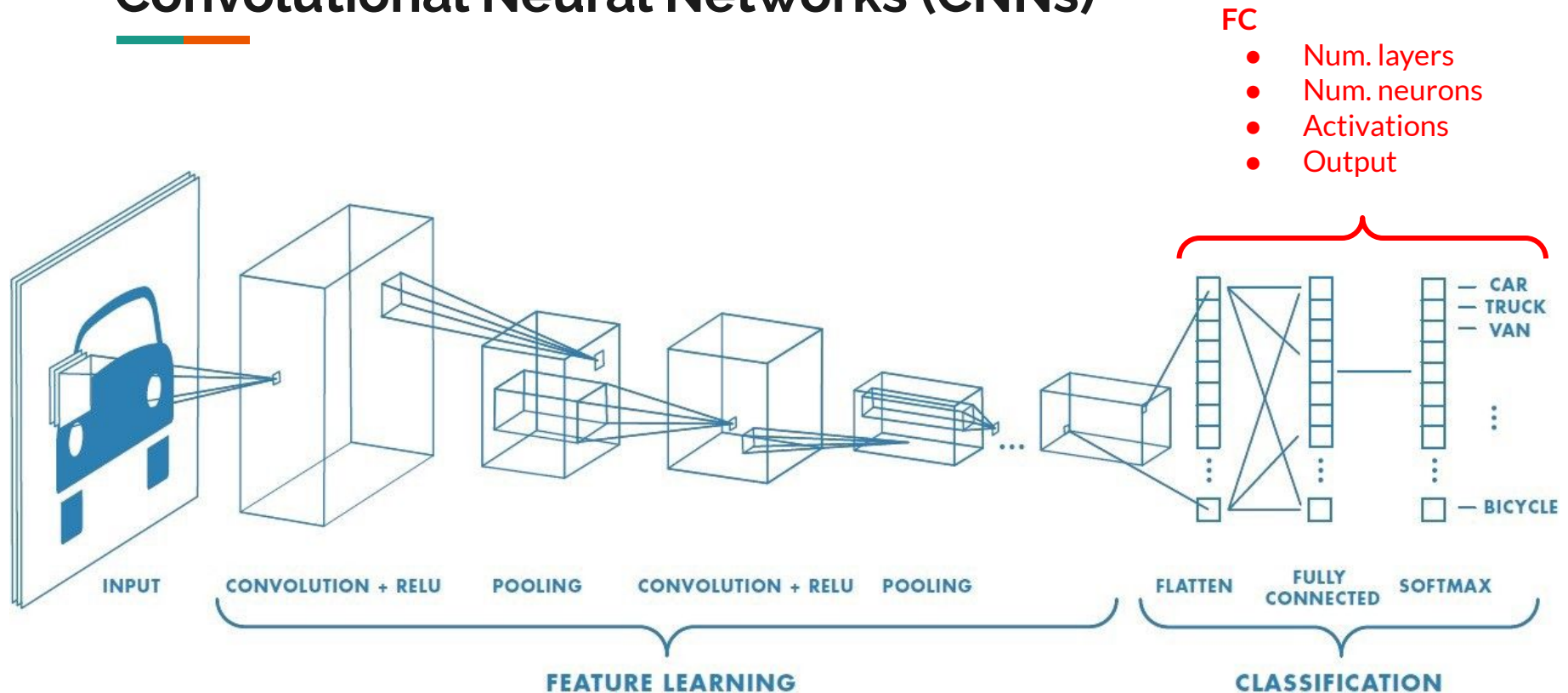
Convolutional Neural Networks (CNNs)



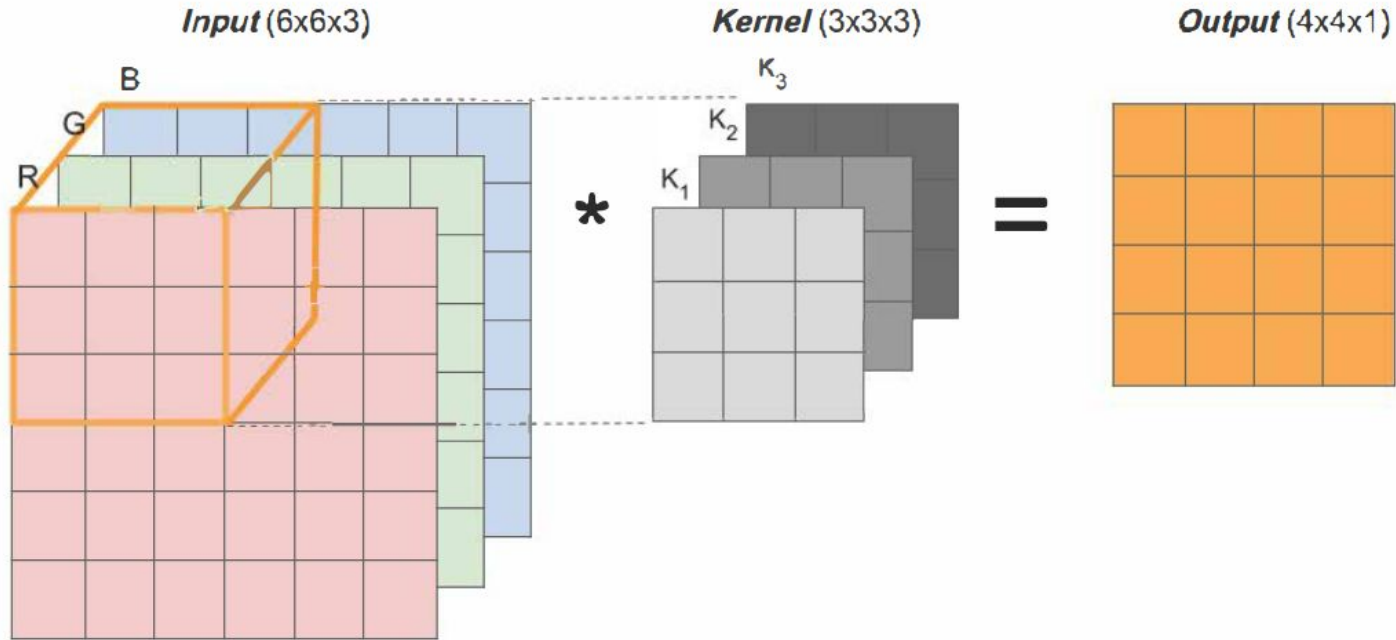
Convolutional Neural Networks (CNNs)



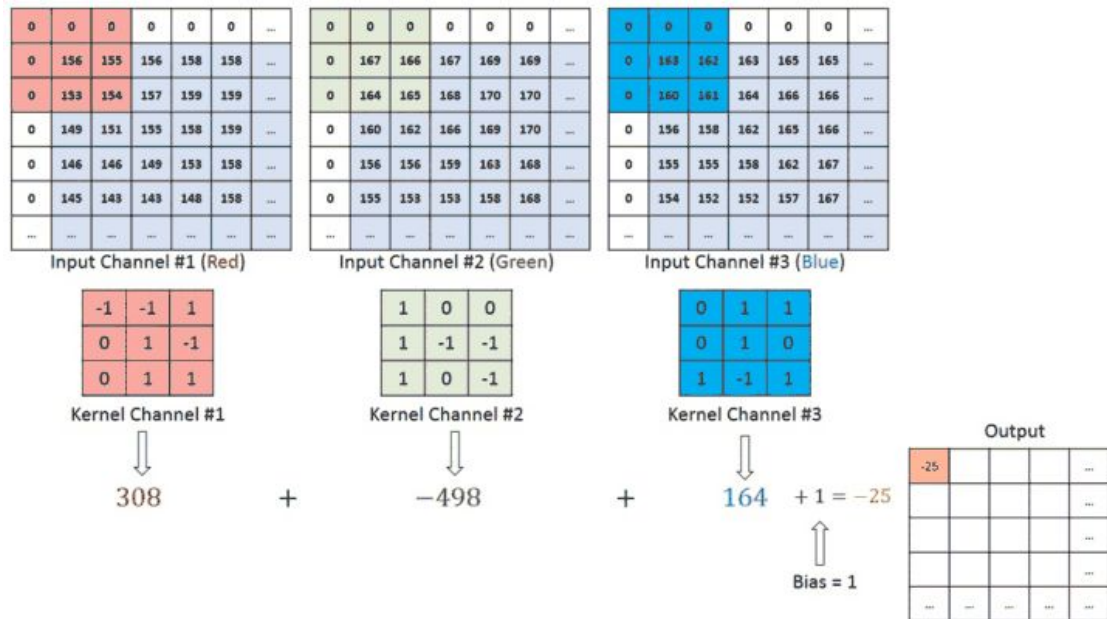
Convolutional Neural Networks (CNNs)



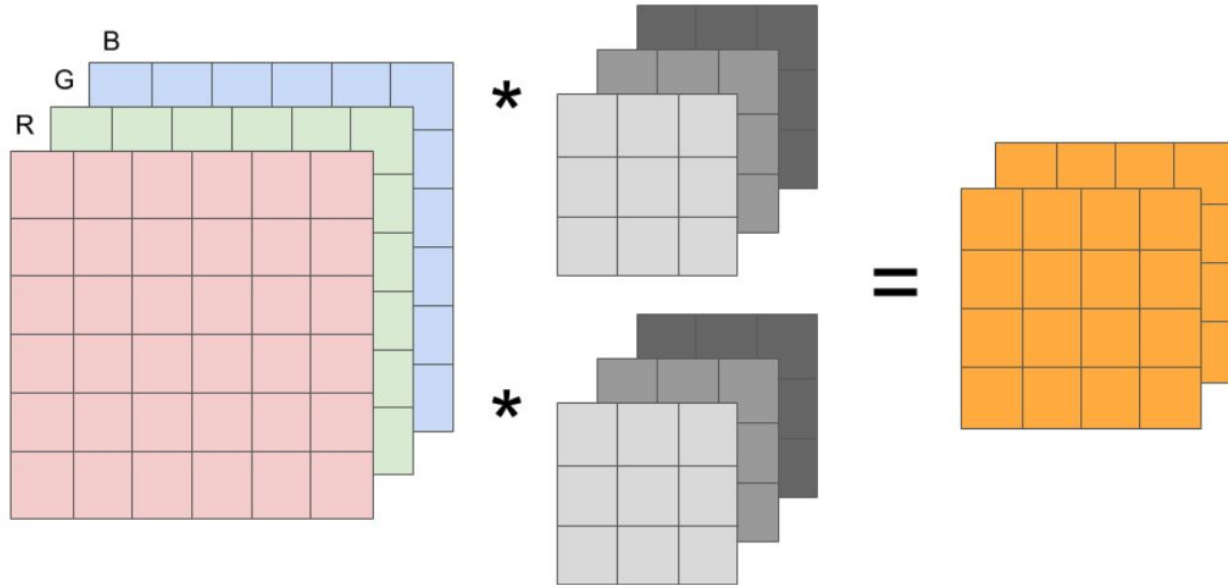
RGB Convolution



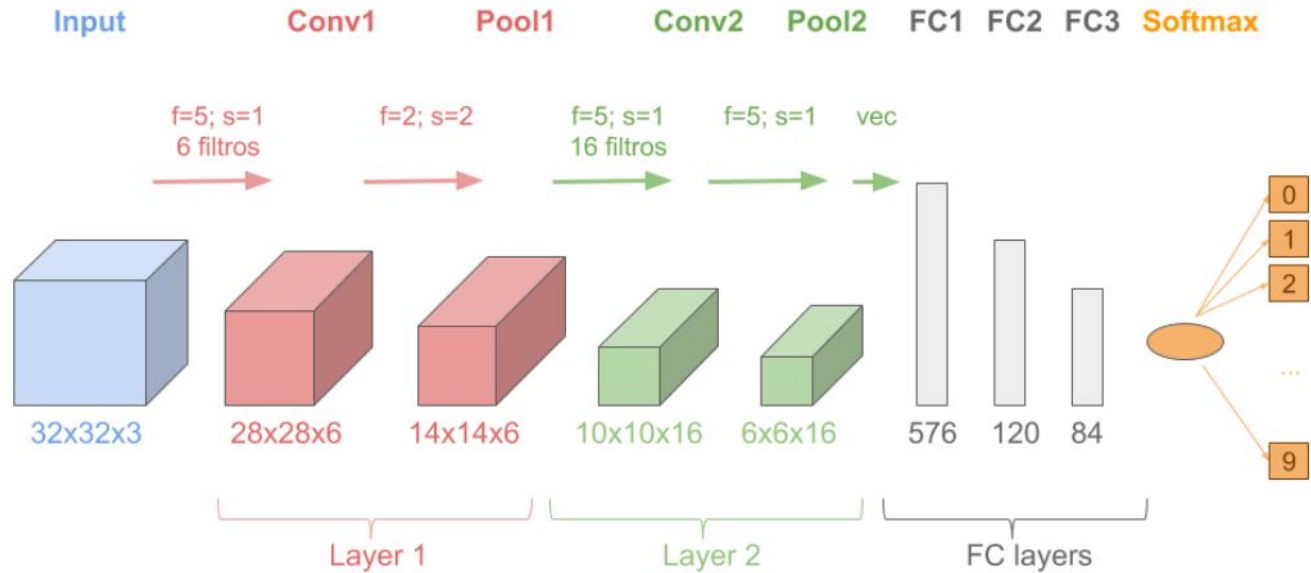
RGB Convolution



RGB Convolution



Inspired in LeNet-5



Fuente: LeCun y col. (1989)

Famous CNN architectures



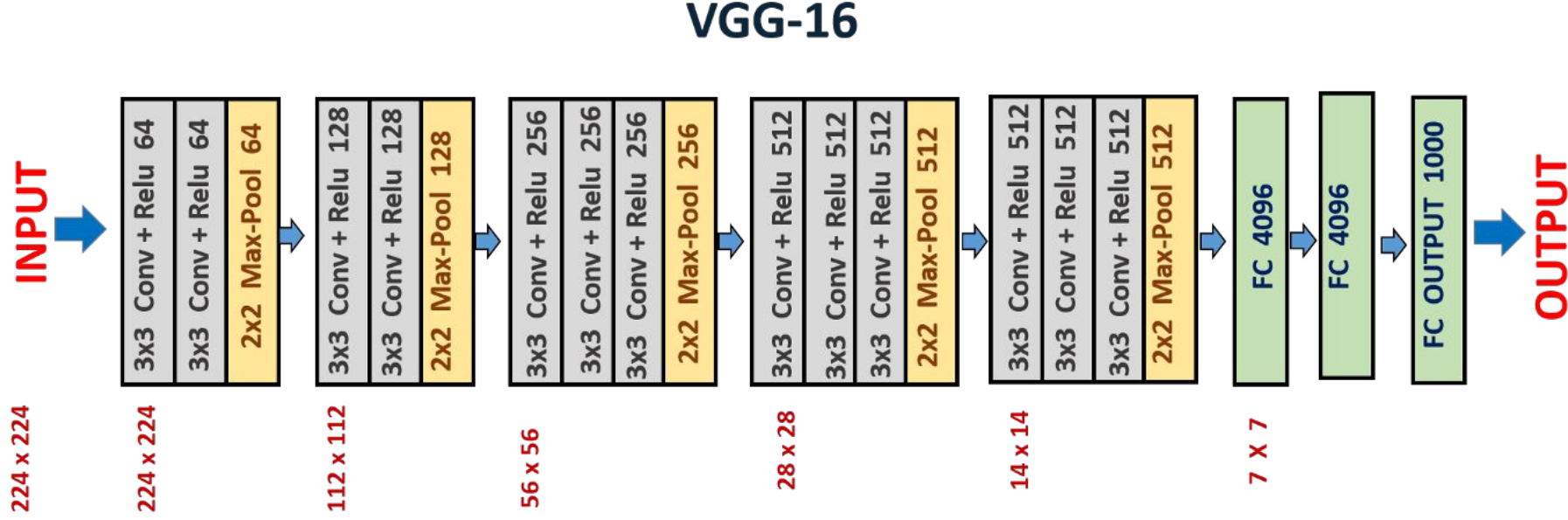
Model name	Number of parameters [Millions]	ImageNet Top 1 Accuracy	Year
AlexNet	60 M	63.3 %	2012
Inception V1	5 M	69.8 %	2014
VGG 16	138 M	74.4 %	2014
VGG 19	144 M	74.5 %	2014
Inception V2	11.2 M	74.8 %	2015
ResNet-50	26 M	77.15 %	2015
ResNet-152	60 M	78.57 %	2015
Inception V3	27 M	78.8 %	2015
DenseNet-121	8 M	74.98 %	2016
DenseNet-264	22M	77.85 %	2016
BiT-L (ResNet)	928 M	87.54 %	2019
NoisyStudent EfficientNet-L2	480 M	88.4 %	2020
Meta Pseudo Labels	480 M	90.2 %	2021

Famous CNN architectures

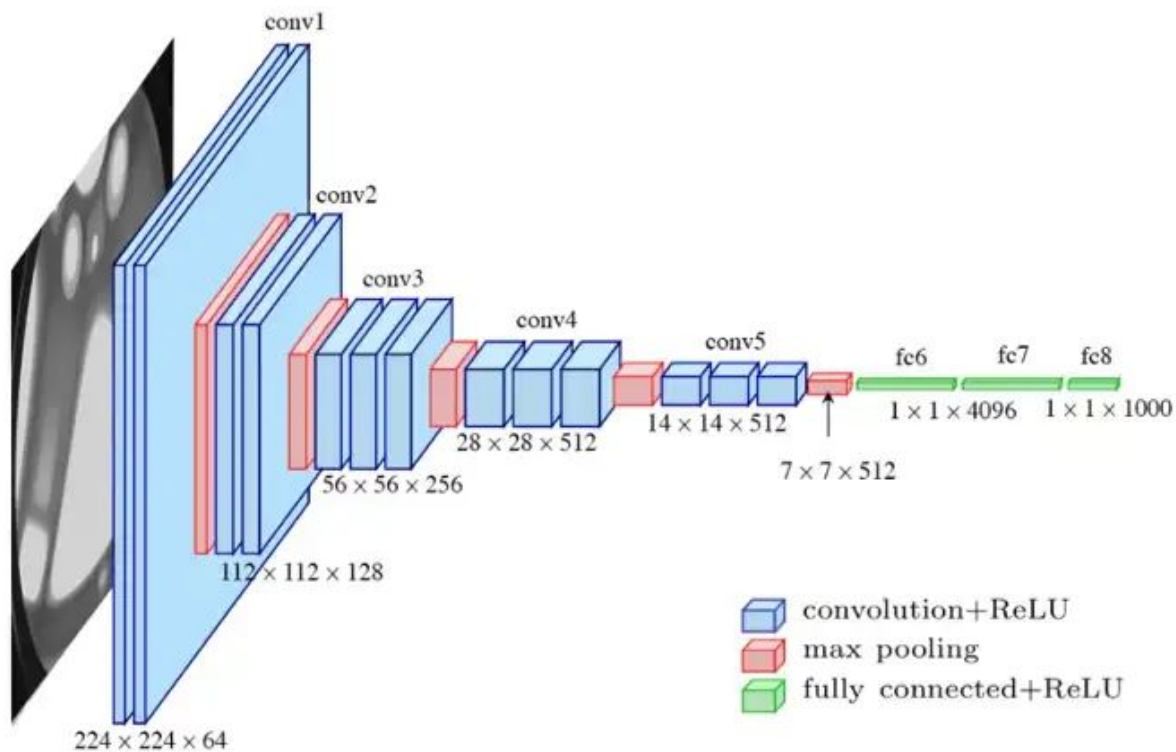


<https://theaisummer.com/cnn-architectures/>

<https://keras.io/api/applications/>



VGG-16





Building CNNs – Recommendations

- Number of kernels
 - Un nombre petit a les primeres capes i anar-lo augmentant a mesura que avances.
 - Capes inicials capturen patrons simples i després els combinen a les capes profundes.
 - Començar amb 32 o 64 filtres és comú.



Building CNNs – Recommendations

- Kernel sizes
 - 3x3 and 5x5 són opcions comunes.
 - Mides més petites van més bé per detalls més específics i petits.



Building CNNs – Recommendations

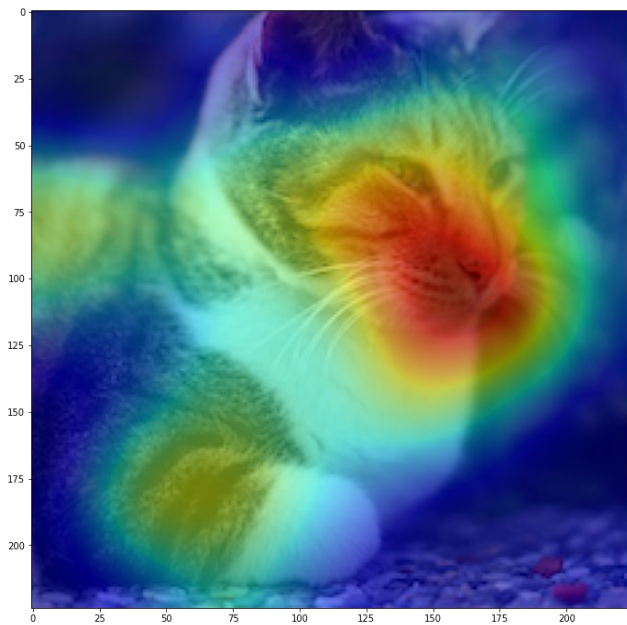
- Pooling layers
 - Generalment utilitzades per reduir dimensió i resumir patrons.
 - Massa pooling pot perdre informació rellevant.
 - 2x2 és molt comú, tot i que algunes tècniques poden arribar a utilitzar 5x5 amb strides de 5.



Building CNNs – Recommendations

- Stride
 - Un stride de 1 és comú en capes convolucionals.
 - Un stride de 2 és comú en capes de pooling.

Interpretability



Misleading classification



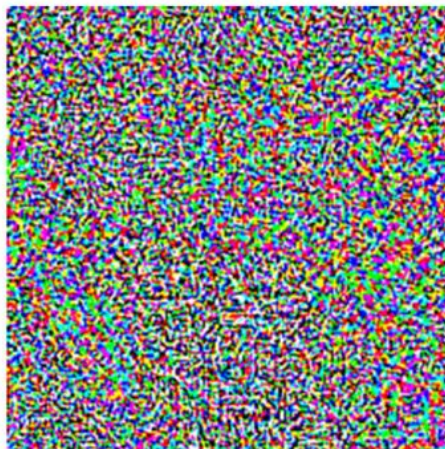
“panda”

57.7% confidence

Misleading classification



+ .007 ×



=



“panda”

57.7% confidence

“gibbon”

99.3% confidence ⁴⁹



Resources

- <https://stanford.edu/~shervine/teaching/cs-230/cheatsheet-convolutional-neural-networks>